Kernel, Frame, Winding

A Geometric Classification of Binary Digit-Extraction Formulas, with BBP as the Octal Member and a Verified Hexagonal Counterpart

Driven by Dean Kulik

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Abstract

The Bailey-Borwein-Plouffe formula is normally treated as a discovered integer relation whose parameters -- the shell spacing 8, the base 16, and the coefficient vector {4, -2, -1, -1} -- are facts about the formula rather than consequences of anything. This paper derives all of them, and shows that BBP is one member of a two-family class rather than an isolated result.

The argument begins at the kernel rather than at the target. The denominator 1 - x^n is the closure condition x^n = 1; its zero set is the n-th roots of unity. The frame is therefore not an interpretation imposed on the formula but the pole structure of its own denominator. Under the eighth root of unity the exponent decomposition 8k + r becomes the quotient map from the integers to the integers modulo 8: complete windings vanish and only the residue survives, as an orientation.

A digit-extraction formula requires an endpoint that is simultaneously the cosine of a rational-degree frame mark and a power of the radix. Sweeping every mark of every regular frame with integer degree measure up to n = 360, and every integer base from 2 to 20, exactly two angles satisfy both conditions: 45 degrees, with cosine 2^(-1/2), and 60 degrees, with cosine 2^(-1). These are the two special right triangles of elementary geometry. They generate two admissible families: octal, with 8 dividing n and base 2^(n/2), and hexagonal, with 6 dividing n and base 2^n.

Two growth operators are distinguished. Refinement doubles the frame and is destroyed at the first step, because the half-angle formula introduces square roots that are not powers of the radix; its convergence is polynomial, dividing the error by four per step. Winding leaves the frame fixed and adds one traversal; each winding contributes exactly one power of the base, so compatibility is preserved indefinitely, and convergence is geometric, dividing the error by sixteen per step. The frame is complete at the first winding: growth occurs in the weight, never in the geometry.

The classification is confirmed by integer relation detection. In the octal frame the standard BBP relation is recovered exactly. In the hexagonal frame, previously unexamined, two relations are found and verified to fifty decimal places: 4*pi*sqrt(3) and 4*pi/sqrt(3), with integer weights over denominators 6k+r at base 64. The constant each frame produces inherits the tangent of its characteristic angle -- tangent 45 degrees is 1, giving clean pi; tangent 60 degrees is sqrt(3), giving pi times sqrt(3). BBP is therefore the octal member of a frame-winding class, not a singular formula.

§1 Definitions

Definition 1 (Frame). A frame of order n is the cyclic set of directions generated by the n-th roots of unity:

$$F_n = \{ \exp(2*pi*i*j/n) : 0 <= j < n \}$$

Each residue class modulo n indexes exactly one element of F_n. The elements are called marks. A mark at angle theta has coordinate cos(theta).

Definition 2 (Kernel). A kernel of order n is the rational function with denominator 1 - x^n. Its poles are exactly the elements of F_n.

Definition 3 (Shell). For a kernel of order n, the shell of index (k, r) is the denominator nk + r appearing in the expansion of the kernel about the origin, with k a non-negative integer and r in {1, ..., n}.

Definition 4 (Endpoint). The endpoint u is the upper limit of integration. A frame-compatible endpoint is one satisfying u = cos(theta) for some mark of F_n. A radix-compatible endpoint additionally satisfies u^n = 1/b for an integer base b.

Definition 5 (Winding and orientation). In the decomposition nk + r, the quotient k is the winding count and the residue r is the orientation.

§2 The Kernel Frame Theorem

Theorem 1 (Kernel Frame). *The frame of a digit-extraction formula is not chosen. It is the zero set of the kernel denominator.*

Proof. The kernel denominator is 1 - x^n. Setting it to zero gives x^n = 1, whose solutions are precisely the n-th roots of unity, which is Definition 1. The poles of the kernel are the frame coordinates. Square.

For the BBP kernel the denominator is 1 - x^8, so n = 8 immediately, with no external geometric assumption. This is the point at which the present treatment differs from the standard account: the geometry is not an interpretation laid over an analytic identity, it is the singularity structure of the identity itself.

Corollary. The denominators 8k + r appearing in BBP are one per marked direction of F_8, indexed by winding.

§3 The Projection Theorem

Theorem 2 (Projection). *Under repeated application of the n-th root of unity, the winding count is annihilated and only the orientation survives.*

Proof. Let omega = exp(2*pi*i/n). Then omega^(nk+r) = (omega^n)^k * omega^r = 1^k * omega^r = omega^r. The map nk + r maps to r is the quotient homomorphism from the integers to the integers modulo n. Square.

Verified numerically for n = 8, r = 1, and k = 0 through 6: the shell indices 1, 9, 17, 25, 33, 41 and 49 are seven distinct magnitudes, and the angle of omega raised to each is 45.00 degrees in every case.

Remark on wording. The magnitude is not destroyed. The map is a quotient, and k remains fully recoverable from the integer; it is simply not used by the operation. The information is projected out by a choice of representation, not lost. Likewise, directions do not rotate -- operators do. The correct statement is that the rotation operator is applied repeatedly and the orientation is invariant under its n-th power. The integers index applications of that operator; they carry operator depth, not quantity.

§4 The Binary Compatibility Theorem

A digit-extraction formula requires an endpoint that is both frame-compatible and radix-compatible. This section establishes that the joint condition is extremely restrictive.

§4.1 The joint condition

Frame compatibility requires u = cos(2*pi*j/n) for some j. Radix compatibility requires u^n = 1/b with b an integer power of 2. Together:

*cos(2*pi*j/n) = 2^(-a/2) for some non-negative half-integer exponent a/2*

Theorem 3 (Binary Compatibility). *Within the class of marks of regular frames with integer degree measure, and integer bases, exactly two angles satisfy the joint condition: 45 degrees and 60 degrees.*

Evidence. An exhaustive sweep was performed over every regular frame of order n with n dividing 360 (so that all marks have integer degree measure), every mark strictly between 0 and 90 degrees, and every integer base from 2 to 20. The results:

angle	cosine	equals	triangle
45 degrees	0.7071067812	2^(-1/2)	45-45-90 (1, 1, sqrt2)
60 degrees	0.5000000000	2^(-1)	30-60-90 (1, sqrt3, 2)

Base 2 returns both angles. Base 4 returns 60 degrees only, being a power of 2. Every other integer base from 3 to 20 returns nothing at all. Changing the radix does not open new frames, because the binding constraint is on the cosine being a pure power, and only these two rational-degree cosines are pure powers of anything.

Scope. This is an exhaustive finite sweep over a stated class, not a proof for all real angles. The class is: marks of regular frames with n dividing 360, integer bases 2 through 20, cosines equal to a half-integer power of the base. A general theorem would require an argument from the theory of cyclotomic fields showing that no other rational-degree cosine lies in the multiplicative group generated by 2. That is stated as an open problem in Section 11.

§4.2 The two families

Each admissible angle generates a family of frames containing it, with the base determined by the endpoint via b = u^(-n):

family	condition	endpoint u	base	first instances
OCTAL	8 divides n	cos 45 = 2^(-1/2)	2^(n/2)	n=8 -> 16, n=16 -> 256, n=24 -> 4096
HEXAGONAL	6 divides n	cos 60 = 2^(-1)	2^n	n=6 -> 64, n=12 -> 4096, n=18 -> 262144

The two families first share a base at 4096, where the octal frame of order 24 and the hexagonal frame of order 12 both apply. Subsequent crossings occur at 2^24 and 2^36.

Retraction. A previous version of this work reported a null result for "base 8 with 6 sectors." That test was malformed: it used the endpoint 2^(-1/2), which is not a mark coordinate of the hexagonal frame, whose coordinates are -1, -0.5, 0.5 and 1. The endpoint did not lie on its own frame. The correct pairing for n = 6 is base 64, and the earlier null is withdrawn as testing an inconsistent object. The distinction matters: it was a failed embedding, not a failed family.

§5 Minimality of the Octal Frame

Two independent conditions each select n = 8 as the minimum octal frame. They are separated here because they are logically distinct.

Theorem 4A (Angular minimality). *The 45-degree direction is a mark of F_n if and only if 8 divides n; the smallest such n is 8.*

Proof. The marks of F_n lie at multiples of 360/n degrees. The angle 45 is such a multiple if and only if 360/n divides 45, equivalently 8 divides n. Square.

n	step	marks (degrees)	45 present
2	180	0, 180	no
4	90	0, 90, 180, 270	no
6	60	0, 60, 120, 180, 240, 300	no
8	45	0, 45, 90, 135, 180, 225, 270, 315	YES

Theorem 4B (Endpoint minimality). *The value 2^(-1/2) is a mark coordinate of F_n if and only if 8 divides n; the smallest such n is 8.*

Proof. Direct computation of the mark coordinate sets. For n = 2 the coordinates are -1 and 1; for n = 4 they are -1, 0 and 1; for n = 6 they are -1, -0.5, 0.5 and 1. In none does 0.7071 appear. For n = 8 the coordinates include plus and minus 0.707107. Square.

Corollary (Minimal octal frame). Both conditions require 8 to divide n. With base b = 2^(n/2), the minimum is n = 8 and b = 16. Base 16 is the smallest base in which 45 degrees exists as a direction. This is the answer to why the formula uses hexadecimal: not convenience for binary hardware, but the minimum frame in which the target angle has a name.

§6 The Weights, Derived

The coefficient vector is normally quoted. It can be computed, and the computation also produces the relation between numerator support and shell index.

Lemma 1 (Monomial shell reduction). *For a kernel of order n with endpoint u,*

*INTEGRAL from 0 to u of x^p / (1 - x^n) dx = u^(p+1) * SUM over k of (u^n)^k / (n*k + (p+1))*

Proof. Expand the kernel as SUM over k of x^(nk) and integrate term by term. Each term contributes u^(nk+p+1)/(nk+p+1). Factor u^(p+1) out. Square.

Corollary 1 (Shell index). *r = p + 1.* The shell index exceeds the numerator power by exactly one, because integrating x^p produces x^(p+1)/(p+1). This is mechanical and explains why the BBP numerator support {0, 3, 4, 5} corresponds to shells {1, 4, 5, 6} with no exceptions.

Corollary 2 (Weight formula). *w_r = a_p * u^r*, where a_p is the coefficient of x^p in the numerator. For the octal frame u^r = 2^(-r/2).

Applied to the standard BBP numerator N(x) = 4*sqrt(2) - 8x^3 - 4*sqrt(2)*x^4 - 8x^5:

power p	coefficient a_p	shell r	factor 2^(-r/2)	weight
0	4*sqrt2 = 5.65685425	1	0.707106781	4

power p	coefficient a_p	shell r	factor 2^(-r/2)	weight
3	-8	4	0.250000000	-2
4	-4*sqrt2 = -5.65685425	5	0.176776695	-1
5	-8	6	0.125000000	-1

The recovered vector is {4, -2, -1, -1}, matching BBP exactly, and the weighted sum agrees with pi to 25 decimal places. The weights are not free parameters: they are numerator coefficients scaled by powers of the endpoint, and the endpoint is fixed by Section 4.

§7 The Parity Lemma and the Square Root of Two

Lemma 2 (Parity). *For the octal endpoint u = 2^(-1/2), the factor u^r is rational when r is even and lies in the rational multiples of sqrt(2) when r is odd.*

Proof. u^r = 2^(-r/2). For r even, r/2 is an integer and the value is a rational power of 2. For r odd, r/2 is a half-integer and the value is 2^(-(r-1)/2) times 2^(-1/2), which is a rational multiple of 1/sqrt(2). Square.

Corollary. Since the weights w_r = a_p * u^r are integers, the numerator coefficient a_p must carry a compensating factor of sqrt(2) precisely when r is odd. The placement of sqrt(2) in the numerator polynomial is therefore forced, not stylistic.

shell r	parity	u^r	rational	numerator coefficient	carries sqrt2
1	odd	0.707106781	no	4*sqrt2	YES
4	even	0.250000000	yes	-8	no
5	odd	0.176776695	no	-4*sqrt2	YES
6	even	0.125000000	yes	-8	no

The correspondence is exact. And by Theorem 2, odd r corresponds to angles that are odd multiples of 45 degrees -- the diagonals. So sqrt(2), which is the reciprocal of the diagonal coordinate, appears in the algebra at precisely the terms sitting on diagonal directions. The polynomial carries the geometry.

§8 Orbit Topology: Parity, Invertibility, Rotors and Dampers

The four surviving shells split two and two, and the split admits two equivalent descriptions.

Arithmetically: odd r gives odd moduli 8k+r, coprime to 16 for every k; the multiplicative order of 16 is defined and its orbit returns to 1. Even r gives even moduli sharing a factor with 16 -- specifically 4 for r = 4 and 2 for r = 6 -- so 16 is not invertible, the orbit never returns to 1, and the sequence falls into a short trapped cycle.

Geometrically: odd r corresponds to odd multiples of 45 degrees, the diagonals; even r to multiples of 90 degrees, the axes.

The chain is parity to invertibility to orbit topology, and parity is the bridge.

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                        90  (r=2, axis)
                              |
      135 (r=3)  |  45 (r=1) DIAGONAL, ROTOR, w=+4
                        \    |    /
                         \   |   /
  180 -----------+---+---+----------- 0  (r=8, axis)
              (r=4, AXIS,  /  |  \
              DAMPER,    /   |   \
          w=-2)  225 (r=5) |  315 (r=7)
                  DIAGONAL,    |
      ROTOR, w=-1  270  (r=6, AXIS, DAMPER, w=-1)
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Figure 1. The octal frame F_8. Shells used by BBP are r = 1, 4, 5 and 6. Residues 1 and 5 are 180 degrees apart: they are one line, the diagonal, read from both ends and weighted +4 and -1. Residues 4 and 6 are the negative x and negative y axes. The formula is one diagonal plus two axes, not four unrelated terms. The endpoint u = cos(45 degrees) is the coordinate of the diagonal direction, so the integration proceeds along the rotor line and the axis terms are what it is subtracted against.

r	angle	geometry	parity	gcd with 16	orbit	weight
1	45	DIAGONAL	odd	1	returns to 1	+4
5	225	DIAGONAL	odd	1	returns to 1	-1
4	180	AXIS	even	4	trapped cycle	-2
6	270	AXIS	even	2	trapped cycle	-1

§9 The Angular Contribution Theorem

Theorem 5 (Angular Contribution). *Poles on the real axis contribute logarithms of magnitudes; poles off the real axis contribute arguments. A constant that is fundamentally an angle therefore requires an off-axis pole.*

Argument. Writing N(x)/(1 - x^n) in partial fractions as a sum of terms R_j/(x - omega_j) over the marks, each term integrates to R_j * log(x - omega_j). For a complex argument, log(z) = log|z| + i*arg(z). The real part is a logarithm of a magnitude -- a length. The imaginary part is an argument -- an angle. A pole lying on the real axis has no argument to contribute and can only ever produce a length. Square.

Since pi = 4*arctan(1), reaching it requires a term whose argument is 45 degrees, hence a pole at 45 degrees, hence a frame containing that direction. This is consistent with the integer relation results of Section 10: in the octal frame, log 3 is found on residues 2 and 6, which are the 90 and 270 degree axes, while pi is found on residues 1, 4, 5 and 6, including both diagonals.

Status. This is an interpretive reading of a standard partial fraction argument, not a theorem with a completed proof. It is labelled as such in Section 11.

§10 Two Growth Operators, and the Classification Result

§10.1 Refinement and winding

Definition 6. Refinement is the operation F_n maps to F_2n: more directions, same number of traversals. Winding is the operation k maps to k+1: the same directions, one further traversal.

Theorem 6 (Refinement destroys compatibility). *Refinement removes radix compatibility at the first step, in both families.*

Proof. The half-angle formula gives cos(theta/2) = sqrt((1 + cos theta)/2). Starting from cos 60 = 1/2 = 2^(-1), refinement gives cos 30 = sqrt(3)/2 = 0.8660254038, which is not a power of 2. Starting from cos 45 = 2^(-1/2), refinement gives cos 22.5 = 0.9238795325, which is not a power of 2. In both cases the very first refinement introduces a square root that is not a power of the radix. Square.

Theorem 7 (Winding preserves compatibility). *Winding preserves radix compatibility indefinitely.*

Proof. By Lemma 1, the contribution of winding k is u^(nk) = (u^n)^k = 1/b^k, a power of the base at every step. No new algebraic number is introduced. Square.

Measured convergence, Archimedes refinement against BBP winding:

step	REFINE frame	value	error	WIND k	value	error
1	n=6	3.000000000000	1.42e-01	1	3.133333333333	8.26e-03
5	n=96	3.141031950891	5.61e-04	5	3.141592645460	8.13e-09
10	n=3072	3.141592106043	5.48e-07	10	3.141592653590	1.78e-15

Refinement divides the error by approximately 4 per step -- polynomial, error of order 1/n squared. Winding divides by approximately 16 -- geometric. After the same ten steps the two differ by nine orders of magnitude, and only winding retains random access to individual digits.

Corollary (Frame completion). At k = 0 the shells n*0 + r for r = 1 through n already cover every mark of the frame. The geometry is complete at the first winding. Every subsequent k is the same marks at 1/b the weight, and the winding itself projects to unity by Theorem 2. Nothing is constructed; the aggregate reading sharpens. This inverts the usual model of numerical convergence: rather than more steps producing better structure, complete structure permits better measurement.

§10.2 The classification: BBP is not special

If the frame-winding account is correct, the hexagonal family should carry its own identities. This was tested by integer relation detection at 60-digit working precision, using PSLQ on the basis constants B_r = SUM over k of 1/(b^k (n*k + r)) together with candidate targets.

Octal control (n = 8, base 16). The detector recovers the BBP relation exactly:

pi = 4/(8k+1) - 2/(8k+4) - 1/(8k+5) - 1/(8k+6), over 16^k

and additionally returns relations for log 2, log 3 and log 5 in the same frame, with log 3 supported only on the axis residues 2 and 6.

Hexagonal result (n = 6, base 64). The detector returns two relations, both previously unexamined in this framework, verified to 50 decimal places after clearing denominators:

*4*pi*sqrt(3) = 20/(6k+1) + 6/(6k+2) - 1/(6k+3) - 3/(6k+4) - 1/(6k+5), over 64^k*

*4*pi/sqrt(3) = 12/(6k+1) - 6/(6k+2) - 3/(6k+3) - 3/(6k+4), over 64^k*

relation	frame	base	series value	|error|
4*pi*sqrt3	n=6	64	21.76559237081061420 71	4.3e-50
4*pi/sqrt3	n=6	64	7.2551974569368714023 8	1.1e-50
pi (BBP control)	n=8	16	3.1415926535897932384 6	3.2e-50

The constant inherits the frame tangent. The octal frame has characteristic angle 45 degrees, whose tangent is 1, so arctan(1) = pi/4 and the constant is pi with no algebraic residue. The hexagonal frame has characteristic angle 60 degrees, whose tangent is sqrt(3), so arctan(sqrt(3)) = pi/3 and the constants carry sqrt(3). The square root appearing in each family is the tangent of its own characteristic direction.

Conclusion. BBP is the octal member of a frame-winding class. The hexagonal member exists, was found by the same method, and is verified. The formula is not singular.

§11 Epistemic Status

Following the type distinction developed in the accompanying session record, claims are sorted by what they contribute. Class I statements introduce a coordinate relation without reducing the admissible set. Class II statements remove a boundary without reducing dimension. Class III statements remove a degree of freedom. Only Class III contributes rank.

Claim	Class	Status
Frame = kernel zero set (Theorem 1)	I	[V] proved
Winding annihilated by projection (Theorem 2)	I	[V] proved, verified k=0..6
Only 45 and 60 degrees are binary-compatible	III	[M] exhaustive over stated class
Angular minimality: 45 first at n=8 (4A)	III	[V] proved
Endpoint minimality: cos45 first at n=8 (4B)	III	[V] proved
r = p+1 (Corollary 1)	I	[V] proved
w_r = a_p u^r, giving {4,-2,-1,-1}	I	[V] computed, 25 digits
Parity Lemma: sqrt2 on odd shells only	II	[V] proved
Parity to invertibility to orbit topology	II	[V] computed
Refinement destroys compatibility (Theorem 6)	III	[V] proved, both families

Claim	Class	Status
Winding preserves it (Theorem 7)	III	[V] proved
Hexagonal relations for 4*pi*sqrt3, 4*pi/sqrt3	III	[V] PSLQ, verified 50 digits
Angular Contribution (Theorem 5)	I	[I] interpretive
Base-8-with-6-sectors null (previous version)	--	[R] RETRACTED, malformed test

[V] verified by proof or computation. [M] measured exhaustively over a stated finite class. [I] interpretive. [R] retracted.

On minimality versus uniqueness of the weights. The constraint on the numerator is a single equation in as many unknowns as there are shells. Over the reals it has infinitely many solutions, so the numerator polynomial is not unique. What is established is that PSLQ, which performs lattice reduction and returns the shortest relation, returns {4, -2, -1, -1}; and that a box search over 28,561 small-integer vectors returns only that vector and its doubling. The relation is minimal, not unique. This also explains why the coefficients are small: short vectors have small coefficients, and "minimal" and "what PSLQ found" are the same statement.

On the discovery machine. PSLQ is blind to the frame. It receives a list of real numbers and returns a short integer combination; it has no representation of angles. The geometry resides entirely in the basis constants B_r, each of which is the accumulated contribution of one marked direction. The machine that discovered BBP never knew it was reading a frame.

§12 Open Problems

Omega 1 (General binary compatibility). Theorem 3 is an exhaustive sweep over a finite class, not a theorem for all real angles. A complete result would use the theory of cyclotomic fields to determine which rational-degree cosines lie in the multiplicative group generated by 2. The conjecture is that only cos 45 and cos 60 do.

Omega 2 (Higher members of each family). The octal family continues at n = 16 with base 256 and n = 24 with base 4096; the hexagonal at n = 12 with base 4096 and n = 18 with base 262144. Whether these carry their own relations, and whether their coefficient vectors bear a systematic relation to the minimal members, is unexamined.

Omega 3 (Crossing bases). The families first share a base at 4096, where the octal frame of order 24 and the hexagonal frame of order 12 both apply, and again at 2^24 and 2^36. A box search restricted to six of the twenty-four available residues at base 4096 returned nothing;

this is a null over a small slice of the search space and should not be read as an absence. Whether a relation exists that uses both geometries at a shared rate is open.

Omega 4 (Other constants, other frames). The octal frame yields relations for log 2, log 3 and log 5; the hexagonal for log 3 and log 7. A systematic classification of which constants appear in which frame, indexed by the angles their defining inverse tangents require, has not been attempted. The conjecture suggested by Theorem 5 is that a constant is reachable in F_n only if it is expressible using angles that are multiples of 360/n.

Omega 5 (The winding coordinate). Theorem 2 shows the winding count is projected out. It is recoverable from the shell index but unused. Whether any extraction scheme is sensitive to k rather than only to r would constitute a different class of construction than the one considered here.

§13 Master Theorem

Theorem 8 (Kernel-Frame-Winding). *Given a kernel of order n, an endpoint that is both a mark coordinate of F_n and a half-integer power of an integer base, and the requirement of positional digit extraction, the following are determined without further choice: the frame, as the kernel zero set; the shell denominators, as one per mark indexed by winding; the shell index, as one more than the numerator power; the coefficient weights, as numerator coefficients scaled by powers of the endpoint; the algebraic support of the numerator, by the parity of the shell index; the orbit topology of each shell, by the parity of its modulus; and the necessity of infinitely many windings, since refinement would destroy compatibility while winding preserves it.*

Within the class swept, exactly two endpoints satisfy the hypothesis: cos 45 degrees and cos 60 degrees. The minimal frames containing them are F_8 with base 16 and F_6 with base 64, and both are shown to carry relations for pi.

The integers appearing throughout are not magnitudes. The quotient records completed windings and is projected away; the residue selects an orientation. They carry traversal state rather than value. The formula is not a computation performed on numbers; it is a reading taken from a frame that was closed before the first term was written.

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